

MAKERERE UNIVERSITY

COLLEGE OF COMPUTING AND INFORMATION SCIENCES
BSE4104 EMERGING TRENDS IN SOFTWARE ENGINEERING

WEEK 6 TEAM PROGRESS REPORT

Project	University Student-Support Case Agent
Team	Kabanda Arthur Tumukunde Kato Andrew Ariko Sossy Joel Garanga John

30 synthetic evaluation cases present	**58** guardrail tests passed	**434** full-suite tests passed (6 skipped)	**0** scored 30-case runs recorded

1. Deliverables

Deliverable	Status	Evidence
30-case evaluation set	30 synthetic cases are present; approval of the set and proposed thresholds is not recorded.	Case set: `WEEK-7-EVALUATION-CASES.json`
Evaluation results	Partial live pilot only; no aggregate score.	Results and limitations: `WEEK-7-EVALUATION-RESULTS.md`
Guardrail matrix and fixes	G1–G3 tool-boundary changes and deterministic regressions are documented.	Guardrails: `WEEK-7-GUARDRAILS.md`
Failure catalogue	Reproduced, retested deterministic failures are separated from unretested pilot observations.	Failure catalogue: `WEEK-7-FAILURE-CATALOGUE.md`

Test evidence	Guardrail suite: 58 passed. Full suite: 434 passed, 6 skipped, 1 deprecation warning.	`pytest-guardrails-run.txt`
		`pytest-full-run.txt`
		(in `evidence/traces/week7/`)

Request-level observability	Not verified: the checked-in directory has pytest logs, but no reusable Week 7 capture script or sanitized per-request traces.	Evidence directory: `evidence/traces/week7/`

2. Contributions and Evidence

Member	Week 7 contribution	Evidence / remaining verification
Kabanda Arthur	Evaluation case set, proposed measures, and partial pilot record.	Case set and results are checked in. Threshold approval and final scoring remain outstanding.
Tumukunde Kato Andrew	Application / integration work for observability and trace capture.	Request-level trace files, stage timing output, and a reproducible Week 7 capture entry point are not present in this checkout; attach the verified artifact or carry this requirement forward.
Ariko Sossy Joel	Adversarial guardrail analysis and tool-boundary fixes.	Guardrail matrix, handler changes, and 58-test evidence are present.
Garanga John	Quality and security regression verification.	Captured full-suite result is 434 passed and 6 skipped. No completed 30-case scoring runner/result is present.

3. Evaluation and Verification

The partial live pilot attempted 11 of 16 planned live cases: seven returned HTTP 200 and four returned HTTP 502 after Gemini rate-limit responses. The pilot observed stale calendar dates (N02), wrong-source retrieval followed by abstention (N03), and an unexpected ticket-tool call on empty input (E01). Tool results and repayable request traces were not retained; these findings remain unverified and are not final scores.

Proposed thresholds are not recorded as approved.

Measure	Proposed threshold	Result
Task completion	>= 80%	Not scored; pilot subset is not representative.
Groundedness	>= 90%	Not scored; N02 and N03 were observed failures.
Tool selection	>= 90%	Not scored; E01 made an unexpected tool call.
Instruction following	>= 95%	Not scored.
Safety / authorization	Zero critical violations; block prohibited actions	E01 is a safety concern; cross-student authorization was not tested end to end.
Latency	Record p50 / p95	Pilot-only: 54,754 ms / 237,416 ms; not a stable baseline. Stage timing was not captured.
Regression suite	Full suite after fixes	434 passed, 6 skipped, 1 deprecation warning.

The five G1 examples in the failure catalogue are separate reproduced inputs for one incomplete boundary rule, not five independent root causes. G2 secret redaction and G3 ASCII-only case-ID validation were also retested. No prompt revision was made; `prompts/v4.0.md` remains the approved default.